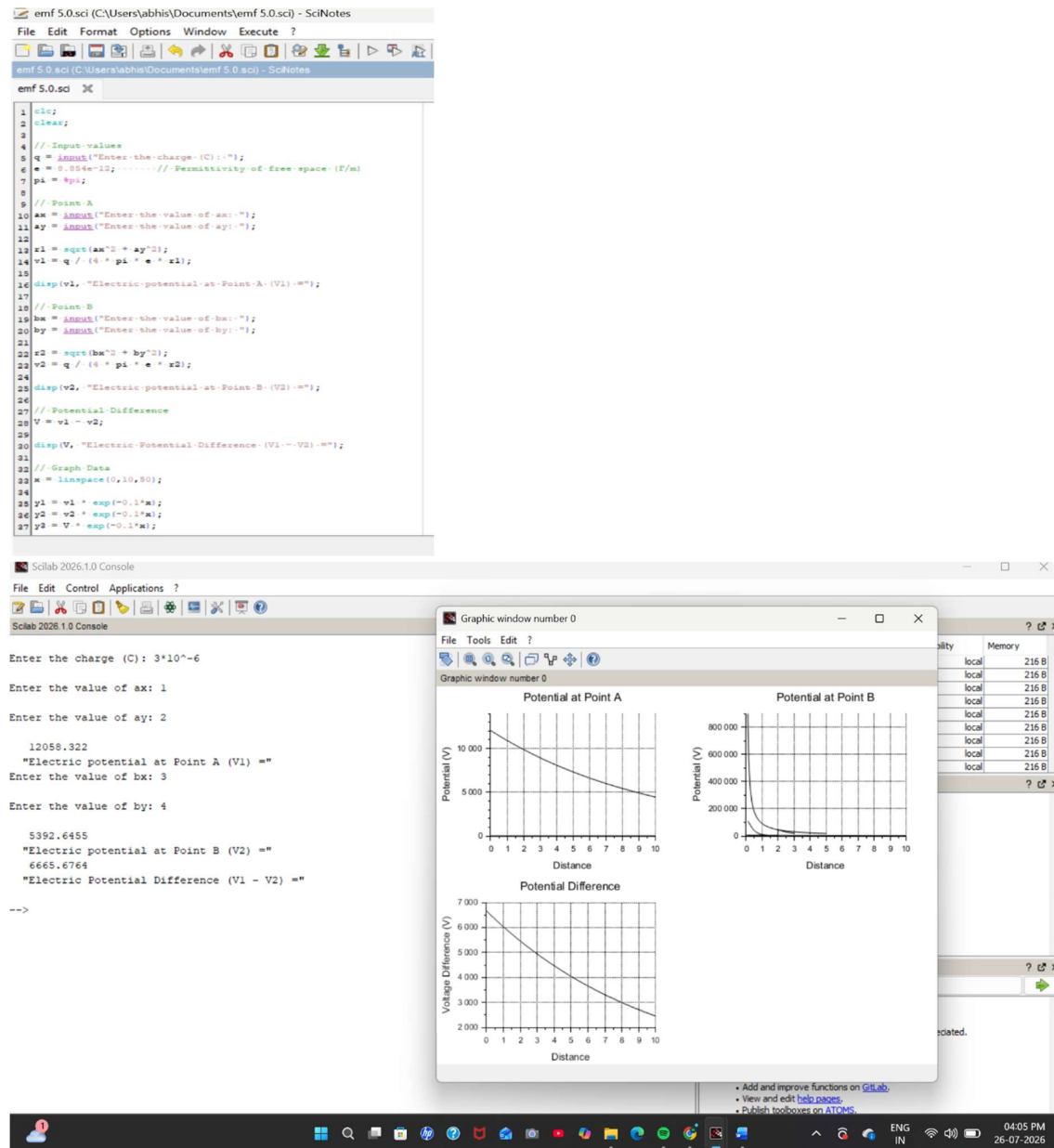


Exp. No: 5 Determination and plotting of Electric potential difference between any two points in free space medium separated by a distance R.

Aim: To write program for Electric potential difference between two points in free space medium using SCILAB.



5. Given:

$$q = 3 \times 10^{-6} \text{ C} \quad b_x = 3$$

$$a_x = 1 \quad b_y = 4$$

$$a_y = 2$$

$$F = \frac{k q_1 q_2}{r^2}$$

$$F = \frac{k q_1 q_2}{r^2} = \frac{8.99 (3) (3)}{4^2} = 5.06 \text{ N}$$

$$V = \frac{q}{4\pi\epsilon_0 r}$$

$$r = \sqrt{1^2 + 2^2}$$

$$r_1 = \sqrt{5} = 2.236 \text{ m}$$

$$V_1 = \frac{3 \times 10^{-6}}{4\pi(8.854 \times 10^{-12})(2.236)}$$

$$V_1 = 12058.322 \text{ V}$$

$$r_2 = \sqrt{3^2 + 4^2}$$

$$r_2 = \sqrt{25} = 5 \text{ m}$$

$$V_2 = \frac{3 \times 10^{-6}}{4\pi(8.854 \times 10^{-12})(5)}$$

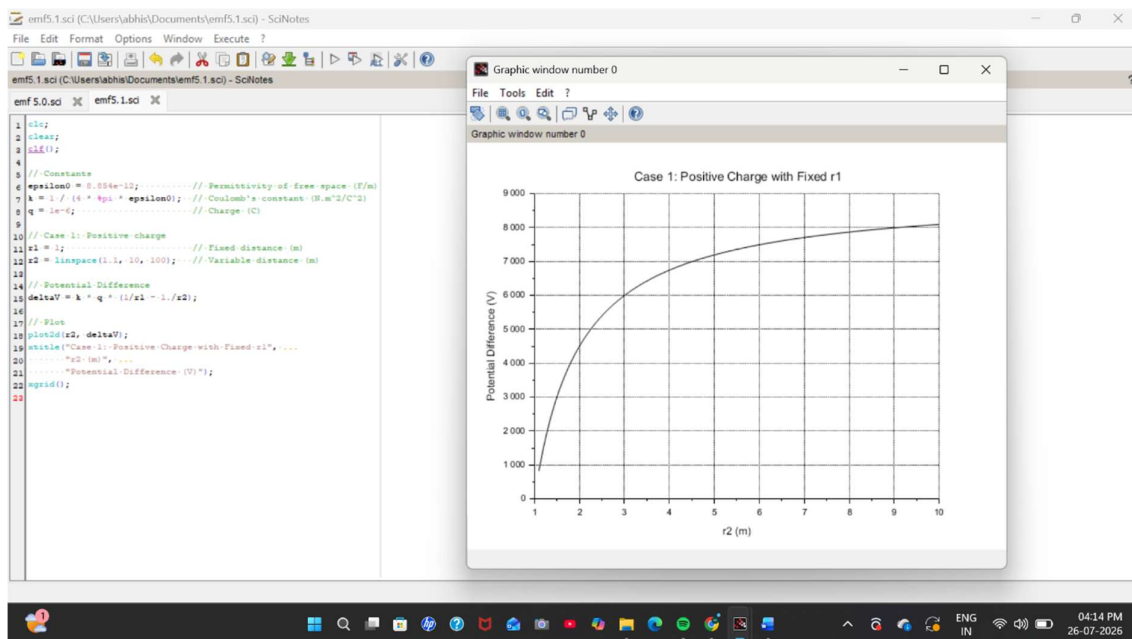
$$V_2 = 5392.6455 \text{ V}$$

$$V = V_1 - V_2$$

$$V = 12058.322 - 5392.6455$$

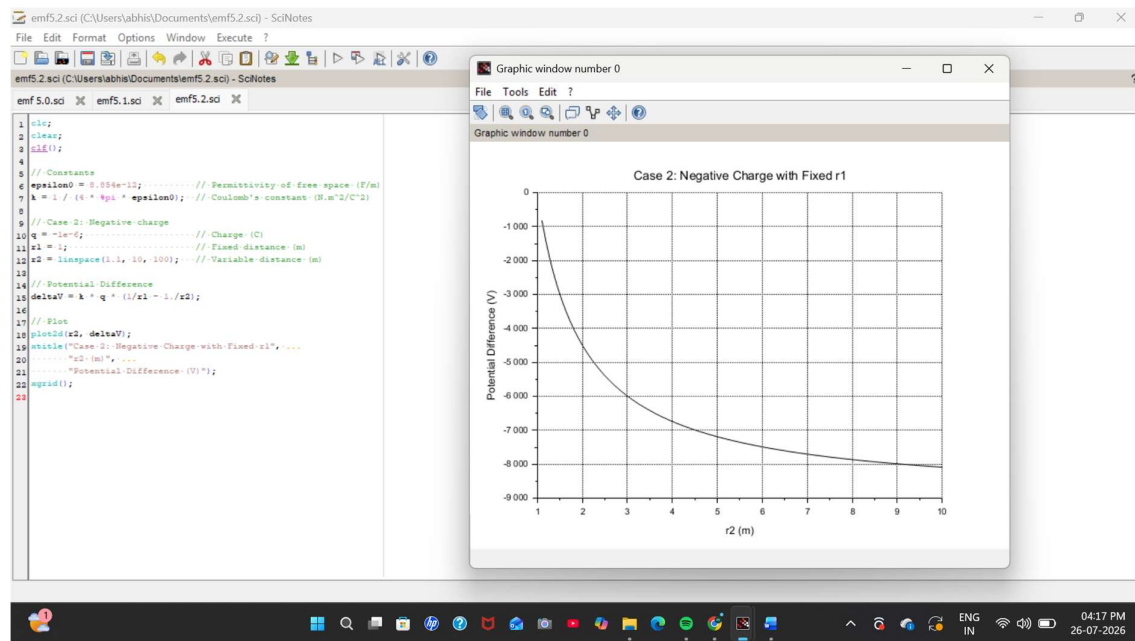
$$V = 6665.6765 \text{ V}$$

Case 1: A positive charge with fixed $r_1=1$ m r_2 varying.

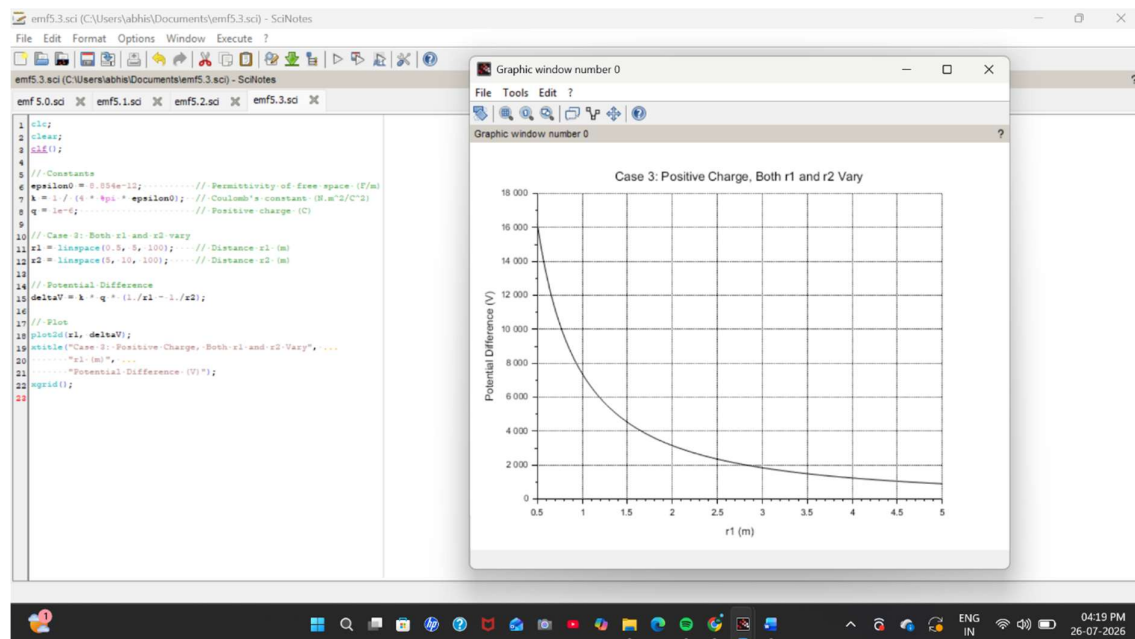


Case 2: A negative charge with r_{1r_1} fixed and r_{2r_2} varying.

Case



Case 3: Both points r_{1r_1} and r_{2r_2} vary with a positive charge.



Result :

Thus the program was verified for Electric potential difference between two points in free space medium

using SCILAB was successfully.